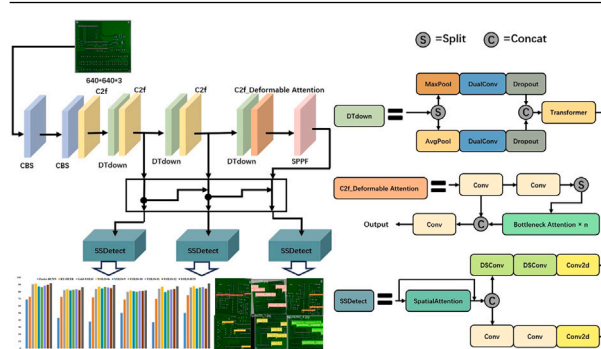


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GRAPHICAL ABSTRACT

- Propose YOLO-DTS for PCB defect detection in complex environments.
- Construct a dual-branch structure with Transformer to enhance feature extraction.
- Combine depth-wise separable convolution and improved attention to boost performance.
- Propose a new loss function combining IoU and NWD to enhance model generalization.
- Achieve high-precision results with lower computational load on public datasets.



ABSTRACT

Keywords:
AI in engineering
Defect detection
Spatial attention mechanism
Deep learning
Computer vision
Transformer

The integrity of Printed Circuit Boards (PCBs) is critical to the performance and reliability of electronic devices. However, existing deep learning-based detection methods often struggle to accurately identify small and complex defects in challenging industrial environments, primarily due to their inability to effectively model irregular defect morphologies, sensitivity to background noise, insufficient multi-scale feature fusion, and difficulties in achieving efficient detection under limited computational resources. To address these issues, we propose a novel object detection framework based on YOLOv8, which integrates a dual-branch feature extraction module, a deformable attention mechanism, and an enhanced spatial attention head. Specifically, we design a Dual-Transformer downsampling module that effectively captures both global context and local details of PCB defects. We also introduce a deformable attention mechanism into the C2f module to handle irregular defect shapes adaptively. Furthermore, we propose a lightweight detection head that employs multi-scale spatial attention and depthwise separable convolutions to enhance feature representation while reducing computational cost. To improve the localization accuracy of small defects, we introduce a new loss function that combines Normalized Wasserstein Distance with Wise-IoUv3. Extensive experiments on the publicly available PKU-Market-PCB dataset demonstrate that YOLO-DTS achieves a precision of 88.4 %, a recall of 69.9 %, and an mAP_{50} of 77.5 %, outperforming the baseline YOLOv8 by 4.9 %, 7.5 %, and 7.5 %, respectively. The parameters used have been reduced by 13.4 % compared

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to the baseline model. Additional experiments on DeepPCB and aluminum profile defect datasets further validate the strong generalization capability of our method. The results indicate that YOLO-DTS is a robust and efficient solution for PCB defect detection in real-world industrial scenarios.

1. Introduction

In today's rapidly advancing technological landscape, Artificial Intelligence (AI) is driving transformation across multiple industries. AI servers, as key infrastructure, are essential for supporting AI applications and development, and the demand for them is continuously growing. Printed circuit boards (PCBs), which are essential components for advanced chips, must maintain excellent electrical performance and stability. Even the slightest defect in PCBs can impact chip functionality, potentially leading to failure. Therefore, PCB defect detection plays a critical role in ensuring proper production quality [1].

With the rapid advancement of deep learning, visual inspection methods now automatically extract more distinct features [2]. The widespread adoption of deep learning-based visual inspection has significantly impacted various industrial applications [3]. These methods are primarily categorized into single-stage and two-stage algorithms [4]. Two-stage algorithms, such as Faster R-CNN, first generate candidate regions and then classify the targets, followed by precise localization. In contrast, single-stage algorithms simplify the process by eliminating the generation of candidate regions, directly predicting both the object classes and bounding box positions. Notably, the YOLO series excels in speed and accuracy, delivering real-time performance with outstanding results [5]. Recently, Wang et al. [6] enhanced detection efficiency and localization accuracy by optimizing the backbone network and fine-tuning the loss function. Yuan et al. [7] enhanced small defect detection by integrating a high-resolution P2 feature layer and strided convolution-free operations. Wu et al. [8] employed cross-scale convolutional fusion with window-based attention to enhance multi-scale defect feature representation.

Despite these advancements, PCB defect detection faces escalating challenges. The rapid development of semiconductor manufacturing processes has led to surface defects becoming smaller and more complex [9,10]. Additionally, environmental factors—such as poor lighting, camera malfunctions, and complex backgrounds—pose further difficulties [11]. While deep learning methods have improved, existing YOLO-based approaches often struggle with these complexities: they lack specialized modules to capture fine-grained features of micro-defects, are vulnerable to background noise, and exhibit poor generalization under varying imaging conditions [12].

To address these limitations, this paper proposes a novel deep learning-based PCB defect detection method named YOLO-DTS. Our approach integrates dual-branch feature extraction, a Transformer model, and an improved spatial attention mechanism to enhance feature representation capabilities. Specifically, a new downsampling strategy is designed to effectively capture features from small and complex defects. A deformable attention mechanism is incorporated to enhance model flexibility, allowing it to adapt to shape variations in target regions. Furthermore, the Neck network fuses multi-scale features, and a new detection head extracts key features for more accurate predictions.

The main contributions of this paper include:

- We propose a feature extraction module that combines dual-branch feature extraction with a Transformer architecture (DTdown). By incorporating grouped convolutions and the Transformer mechanism, this module is more effective at capturing contextual information. Furthermore, a Deformable Attention mechanism is integrated into the Convolution to Focus (C2f) module, significantly improving the model's ability to extract local features.
- We design a lightweight detection head that integrates an enhanced spatial attention mechanism with depth-wise separable convolutions (SDDetect). The original single convolution in the spatial attention

mechanism is replaced with three independent convolutions, which not only expand the receptive field but also reduce the number of parameters, thereby improving performance.

- To improve the accuracy of small target detection and mitigate the impact of small target position shifts, we introduce a novel loss function: Normalized Wasserstein Distance-Weighted Intersection over Union version 3 (NWD-WIoUv3). This metric addresses the sensitivity of traditional Intersection over Union to small position shifts of targets, thus enhancing the model's generalization capability.

This paper presents an efficient method for detecting small PCB defects in complex industrial environments. The proposed model effectively handles various real-world disturbances, such as insufficient lighting and electromagnetic interference, while optimizing computational resource usage, thereby significantly reducing the computational burden. With its high processing efficiency and outstanding detection accuracy, the method enables precise defect identification even in resource-constrained settings.

The structure of this paper is organized as follows: [Section 2](#) reviews the progress of PCB defect detection research using artificial intelligence techniques. [Section 3](#) explains the YOLO-DTS model and its innovations. [Section 4](#) describes the characteristics of the public PCB defect datasets, data augmentation strategies, and the experimental setup. [Section 5](#) presents the validation of the proposed method, comparing its performance with state-of-the-art algorithms on the same test datasets. Finally, [Section 6](#) concludes the paper and suggests directions for future research.

2. Related works on PCB defect detection techniques

This section reviews the evolution of PCB defect detection techniques, categorizing them into traditional methods, deep learning-based approaches, and recent YOLO variants, while identifying research gaps that our work aims to fill.

2.1. Evolution of PCB defect detection techniques

PCB defect detection is paramount for ensuring product reliability. Early methods leveraged traditional machine learning and image processing to automate optical inspection, yet they struggled with the critical challenge of precise image alignment between test and reference samples [13].

The convergence of Industry 4.0 and deep learning has profoundly advanced object detection, which is broadly categorized into two-stage and single-stage paradigms. Two-stage methods, exemplified by Faster R-CNN, first generate region proposals before performing classification and regression. Liu et al. [14] proposed a two-stage approach adept at preserving features of small defects, thereby enhancing PCB inspection accuracy. Liu et al. [15] further integrated deformable transformers with pyramid feature fusion to boost multi-scale defect detection efficiency. Despite their high accuracy, these methods are often hampered by substantial computational loads and limited real-time capability, constraining their deployment in high-throughput industrial settings.

2.2. The rise of single-stage detectors and YOLO variants

Single-stage detectors streamline the process by directly predicting object classes and bounding boxes, framing detection as a regression problem. This end-to-end design often yields superior inference speed and stronger generalization. Zhang et al. [16] achieved high-precision PCB defect detection using a lightweight feature extraction network coupled with multi-scale aggregation. Li et al. [17] enhanced the Detection

Table 1

Comparison and review of other PCBs defect detection networks and our proposed model.

Ref.	Baseline	Backbone	Detection head	Loss function	Dataset	Precision (%)
Wang et al. [6]	YOLOv7	ELAN-C	Decoupled Detection Head	Siou	PKU-Market-PCB	+ 2.04 %
Yuan et al. [7]	YOLOv8	Conv-SPD + SimAM	Original Detection Head	WIoU	PKU-Market-PCB	+ 0.60 %
Wu et al. [8]	YOLOv12	M2C2f + EAMC	Original Detection Head	–	PKU-Market-PCB	+ 2.30 %
Ge et al. [18]	YOLOX	FEM + AFFM	Original Detection Head	CloU	Ge's dataset	+ 4.20 %
Bai et al. [19]	YOLOv8	ECANet	Small-target Detection Head	CloU	PKU-Market-PCB	+ 4.06 %
Yuan et al. [20]	YOLOv5	HorNet	Optimized Head	CloU	HRIPCB	+ 6.10 %
Xiao et al. [21]	YOLOv7	CA	Decoupled Detection Head	Inner-CloU	PKU-Market-PCB	+ 0.50 %
Zhou et al. [22]	YOLOv5	MobileNetv3 + SCPDarknet53	Decoupled Detection Head	CloU	Augmentation of PKU-Market-PCB	+ 3.34 %
Yu et al. [23]	YOLOv5	GhostNet	C3STR	GloU	PKU-Market-PCB	+ 4.75 %
Our proposed	YOLOv8	DTdown + C2f_DAT	SDDetect	NWD + WIoUv3	Augmentation of PKU-Market-PCB	+ 4.90 %

+ : indicates an enhancement relative to the baseline model.

Transformer (DETR) by incorporating EfficientNet and a spatial attention offset module, improving the recognition of small and complex defects. However, a common limitation persists: these models frequently exhibit performance degradation in complex industrial environments, particularly under challenges such as poor lighting and significant background interference.

The YOLO series, a seminal family of single-stage detectors, is renowned for its exceptional speed-accuracy balance. Its architecture unified classification and localization into a single network, making it highly suitable for real-time applications. Subsequent research has focused on adapting YOLO for PCB inspection: Ge et al. [18] mitigated catastrophic forgetting in incremental learning for YOLOX-S by integrating Feature Enhancement and Attention Feature Fusion modules with Elastic Response Distillation. Bai et al. [19] enhanced YOLOv8 for small defects by incorporating the Efficient Channel Attention network (ECANet) and a dedicated small-target detection head. Yuan et al. [20] improved YOLOv5's feature extraction capabilities by introducing the HorNet structure. Xiao et al. [21] boosted the performance of YOLOv7-tiny using the Coordinate Attention mechanism and Inner-CloU loss. Wang et al. [6] reduced false positives and missed detections in YOLOv7 by designing the ELAN-C module (combining CBAM and ELAN) and employing Siou loss. Zhou et al. [22] integrated MobileNetv3 and CSPDarknet53 with attention mechanisms into YOLOv5, striving for a balance between accuracy and efficiency. YOLO-SSW [7] significantly enhanced the detection accuracy of small targets by introducing a high-resolution P2 feature layer, employing strided convolution-free (Conv-SPD) operations to preserve fine-grained details, and integrating the SimAM parameter-free attention module with the WIoU loss function, achieving superior performance on public datasets. Similarly, YOLO-EMAC [8], built upon YOLOv12, designed a Multiscale and Multibranch Cross-scale Convolutional Fusion (M2C2f) module to mitigate the loss of small target information in deep feature maps and proposed an Efficient-Adaptive-Multibranch-Channel (EAMC) module to adaptively strengthen key features. Yu et al. [23] developed a dedicated lightweight architecture for PCB defect detection by incorporating a GhostNet backbone, depthwise separable convolutions, and a C3STR module integrating Swin Transformer.

2.3. Identifying research gaps and our contributions

Through systematic analysis of existing PCB defect detection methods (as summarized in Table 1), we observe that current research primarily develops along two technical pathways with distinct limitations: The first focuses on improving detection accuracy by introducing complex network architectures, exemplified by YOLO-SSW which enhances small defect detection capability through constructing high-resolution feature pyramids, yet at the cost of sharply increased computational complexity and significantly degraded inference speed. The second pathway emphasizes deployment efficiency through lightweight design, as demonstrated by MSD-YOLO which substantially reduces parameters by integrating MobileNet-v3, but suffers from compromised detection

accuracy in complex backgrounds due to weakened feature extraction capability.

Notably, this “accuracy versus efficiency” trade-off challenge is not unique to PCB inspection. In Unmanned Aerial Vehicle visual tracking tasks, some methods also strive to maintain high precision in complex environments while facing similar computational efficiency challenges [24]. This widespread phenomenon further validates the urgency and importance of developing a solution that balances both accuracy and speed in the field of PCB defect detection.

To break through this bottleneck, we propose YOLO-DTS - an innovative architecture designed to systematically address these intertwined challenges. Compared to existing approaches, our model achieves a breakthrough balance through four core innovations: employing a Dual-branch Transformer downsampling module to ensure multi-scale feature extraction quality while reducing computational costs; embedding a Deformable Attention mechanism within the C2f module to achieve precise focus on irregular defects with minimal computational overhead; designing a lightweight detection head incorporating enhanced spatial attention that maintains superior localization accuracy while significantly reducing parameters; and adopting an NWD-WIoUv3 composite loss function to substantially improve small target detection performance. This comprehensive solution enables YOLO-DTS to achieve detection accuracy comparable to the most advanced accuracy-oriented models while significantly surpassing them in inference efficiency, truly realizing the optimal balance between accuracy and efficiency, thereby providing an ideal solution for industrial deployment.

3. A YOLO-DTS based methodology for PCB defect detection

This section details the architecture of YOLO-DTS, including the DTdown module for multi-scale feature extraction, deformable attention in C2f blocks, the lightweight SDDetect head, and the composite NWD-WIoUv3 loss function. Fig. 1 illustrates the overall architecture of the proposed YOLO-DTS model. It is built upon the YOLOv8 framework but incorporates several key modifications to address the challenges: Specifically: Firstly, we replace the original downsampling modules in the backbone with our Dual-branch Transformer down-sampling module to enhance multi-scale feature extraction. Secondly, we integrate the Deformable Attention mechanism into the C2f modules to form C2f_DAT, enabling adaptive spatial focus. Next, the original detection heads are substituted with our Spatial Attention-DSConv-Detection Head to achieve lightweight and precise prediction. Finally, the NWD-WIoUv3 loss function is introduced to replace the original loss, improving localization accuracy for small defects. The following subsections detail each component.

3.1. C2f module with deformable attention for enhanced feature extraction

The standard self-attention mechanism, while powerful, operates on fixed grid points within a feature map, making it less effective at capturing the irregular and varying shapes of PCB defects. To overcome this limitation, we incorporate the Deformable Attention (DAT) mechanism

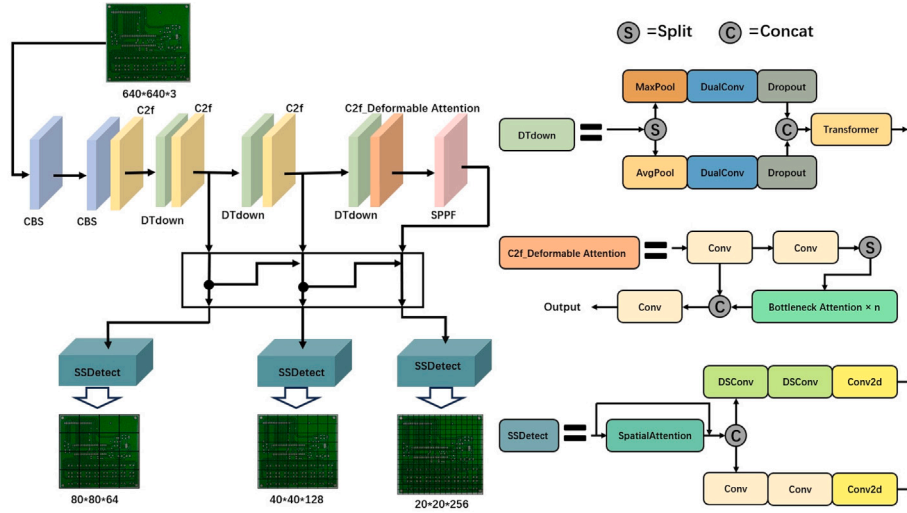


Fig. 1. Overall structure of YOLO-DTS.

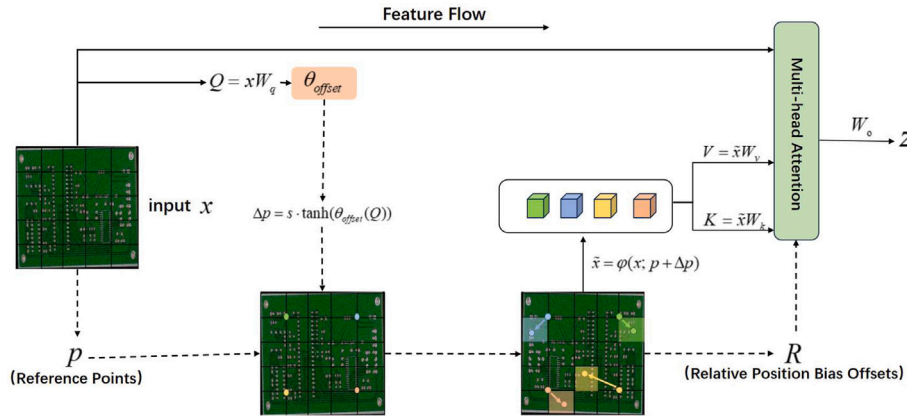


Fig. 2. Deformable attention mechanism.

[25], which introduces learnable offsets to dynamically adjust the sampling locations based on the input features. Its schematic is shown in Fig. 2.

Given an input feature map $x \in R^{H \times W \times C}$, a grid of reference points $p \in R^{H_G \times W_G \times 2}$ is first generated, where $H_G = H/r$, $W_G = W/r$, and r is the down-sampling ratio.

The query embedding Q is obtained through a linear projection of x :

$$Q = xW_q \quad (1)$$

To compute the offset for each reference point, Q is fed into an offset network θ_{offset} . A scaling factor s is introduced to control the magnitude of the offsets and ensure training stability. The final offset Δp is calculated as:

$$\Delta p = s \cdot \tanh(\theta_{offset}(Q)) \quad (2)$$

The $\tanh$ function constrains the initial offsets to the range $[-1, 1]$, which are then scaled by s . The adjusted reference points $p + \Delta p$ are used to sample features from the original input x via bilinear interpolation $\phi(\cdot)$:

$$\tilde{x} = \phi(x; p + \Delta p) \quad (3)$$

This deformed feature map $\tilde{x}$ is then projected to form the Key (K) and Value (V) embeddings using their respective linear projection

matrices W_k and W_v :

$$K = \tilde{x}W_k, V = \tilde{x}W_v \quad (4)$$

The Query (Q), Key (K), and Value (V) are then fed into a multi-head attention mechanism. The output of the m -th head is computed as:

$$z^{(m)} = \sigma \left(Q^{(m)} (K^{(m)})^T / \sqrt{d} \right) V^{(m)}, m = 1, 2, \dots, N \quad (5)$$

where $\sigma(\cdot)$ denotes the softmax function, and d is the feature dimension per head. The outputs of all heads are concatenated and passed through a final linear projection W_o :

$$z = \text{concat}(z^{(1)}, \dots, z^{(N)}) W_o \quad (6)$$

We integrate this DAT mechanism into the YOLOv8 architecture by replacing the standard convolutional layers within the Bottleneck blocks of the original C2f module, as depicted in Fig. 3. The Add represents element-wise addition for residual connection. This integration enables the model to adaptively focus on more informative regions, capturing fine-grained features of small and irregular defects effectively. The residual connection fuses the enhanced features with the original input, strengthening feature representation and facilitating the training process. The proposed C2f-DAT module excels at suppressing background noise and redundant information in complex industrial environments, significantly improving the accuracy and robustness of PCB defect detection.

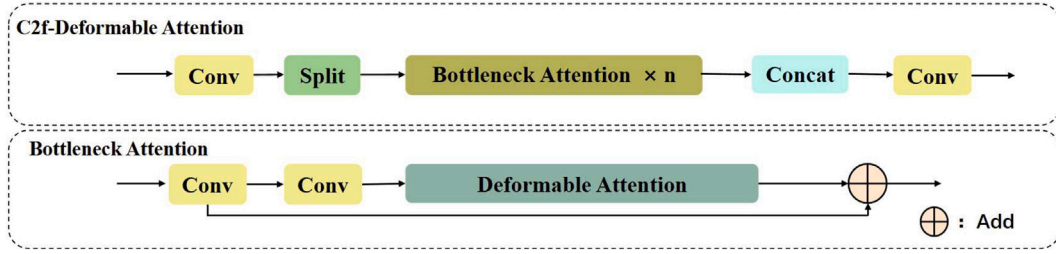


Fig. 3. C2f-Deformable attention integration.

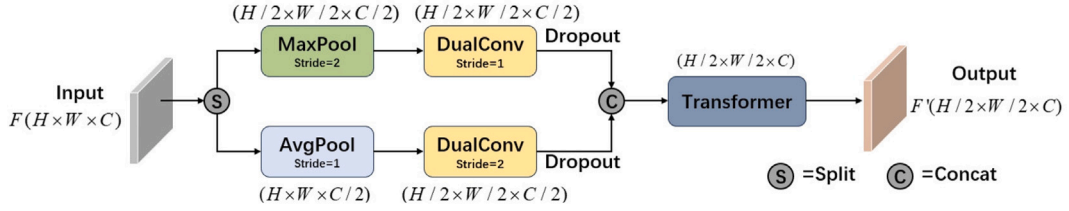


Fig. 4. DTdown module for enhanced feature extraction.

3.2. Dual-branch feature extraction with transformer

Standard down-sampling operations in YOLOv8, such as strided convolution, can lead to the loss of fine-grained features crucial for detecting minute PCB defects and are susceptible to amplifying background noise. Inspired by Adown [26], we propose a Dual-branch Transformer down-sampling module (DTdown) that explicitly decouples the extraction of global contextual information and local detailed features. This design leverages the complementary strengths of both branches to enhance multi-scale representation capabilities.

The DTdown module is built upon a DualConv operation [27]. We define DualConv as the concatenation of Group Convolution (GC) and Pointwise Convolution (PWC) outputs:

$$\text{DualConv}(x) = \text{Concat}(\text{GC}(x), \text{PWC}(x)) \quad (7)$$

where $\text{GC}(\cdot)$ denotes a group convolution with 3×3 kernels, which efficiently extracts spatial features while reducing parameters. $\text{PWC}(\cdot)$ represents a pointwise convolution (1×1), responsible for cross-channel feature fusion and information reuse. This parallel structure maintains feature diversity and enhances local perceptual capability, making it particularly suitable for PCB images characterized by complex textures and fine-grained defects.

The overall architecture of DTdown is illustrated in Fig. 4. The input feature map x is partitioned into two processing pathways: The left branch employs average pooling followed by a DualConv operation. This branch focuses on extracting smooth, global contextual information and background texture features of the PCB substrate, which provide structural prior knowledge. The right branch utilizes max pooling combined with a DualConv operation. This branch is designed to capture salient local detailed features of minute defects, such as sharp edges and irregularities, while inherently suppressing irrelevant background noise commonly encountered in industrial scenarios.

This dual-path design processes overall environmental information and microscopic defect features separately, preventing mutual interference between these two types of information within a single branch. A Dropout operation is incorporated after each convolutional layer to mitigate potential overfitting on the training set and enhance the model's generalization capability in complex industrial environments.

Finally, the output features from both branches are concatenated and fed into a standard Transformer encoder:

$$\text{output} = \text{Transformer}(\text{Concat}(\text{Branch}_{\text{MaxPool}}(x), \text{Branch}_{\text{AvgPool}}(x))) \quad (8)$$

This module establishes long-range dependencies through its self-attention mechanism, which is critical for addressing the dispersion and morphological diversity of PCB defects. Firstly, it can correlate potential defect features across different regions, enhancing the perception of minute and sparse defects; Secondly, it adaptively fuses the global environmental information from the left branch with the local defect features from the right branch, improving semantic understanding of defect contexts; Finally, positional encoding preserves the spatial layout information of PCB components, while the feed-forward networks enhance feature discriminability. This integration significantly improves the detection accuracy and robustness for small defects in complex industrial backgrounds.

3.3. Lightweight detection head based on improved spatial attention and depthwise separable convolution

The original YOLOv8 detection head employs standard convolutions, which entail a significant parameter footprint. Moreover, its lack of a dedicated spatial attention mechanism limits its ability to focus on critical defect regions amidst complex PCB backgrounds. To address these limitations, we propose the Spatial Attention-DSConv-Detection Head (SDDetect), a lightweight and attentive detection head. The structure of SDDetect is shown in Fig. 5. In the diagram, Add denotes element-wise addition for residual connections, Attn denotes attention weighting, and Merge denotes the aggregation of multi-branch features.

Upon receiving the input feature map F , an enhanced spatial attention mechanism is applied to compute attention weights. Typical spatial attention modules often rely on a single 7×7 convolution, which lacks the adaptability to capture defects at multiple scales and has limited sensitivity to fine local details. Instead, we propose a multi-scale parallel design. We use three separate convolutional branches with kernel sizes of 3×3 , 5×5 , and 7×7 to process the input features in parallel: The 3×3 kernel is optimal for capturing local fine-grained details and textures; The 5×5 kernel provides a wider receptive field, integrating context from slightly larger areas; The 7×7 kernel offers the largest receptive field to capture global context and spatial relationships within the feature map.

Comparison with Existing Attention Mechanisms. The proposed spatial attention mechanism differs fundamentally from prevalent designs like CBAM [28] and Coordinate Attention [29]. Unlike CBAM's spatial attention, which employs a single 7×7 convolution, our multi-scale parallel design adaptively aggregates features from varying receptive fields. Furthermore, while Coordinate Attention excels at encoding

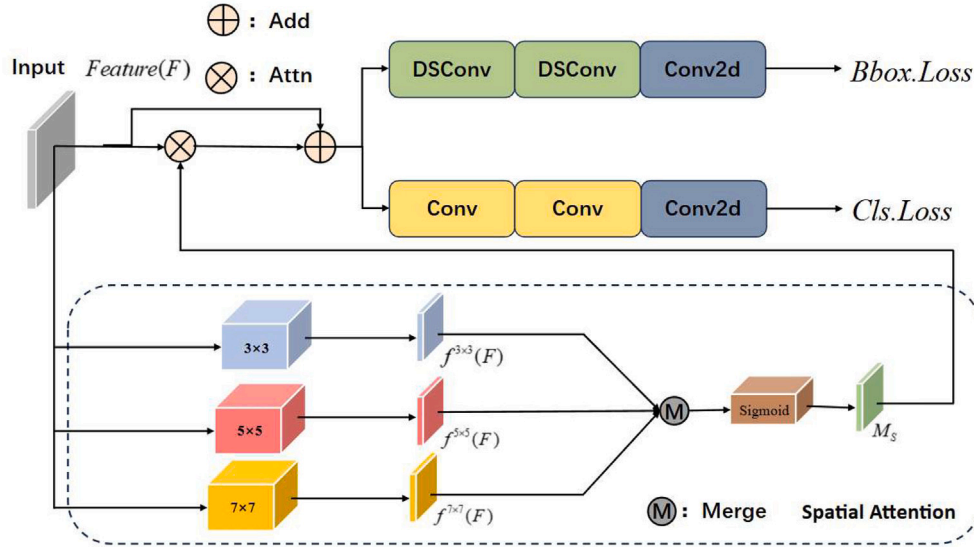


Fig. 5. Detection head architecture with spatial attention and depth-wise separable convolutions.

positional information through channel attention, our method focuses on harnessing rich, multi-scale spatial context for discerning complex defect patterns against cluttered backgrounds, which is a more critical requirement in PCB inspection.

The output feature maps from these three convolutional layers are element-wise summed to aggregate multi-scale information. The summed result is then normalized using a sigmoid activation function to generate the final spatial attention map $M_s(F)$, with values ranging between 0 and 1:

$$M_s(F) = \sigma(f^{3 \times 3}(F) + f^{5 \times 5}(F) + f^{7 \times 7}(F)) \quad (9)$$

The refined feature map F' is obtained by multiplying the original input features F with the attention map $M_s(F)$:

$$F' = F \cdot M_s(F) \quad (10)$$

A residual connection is introduced to preserve the original features, prevent information loss, and ensure stable gradient flow during training. The refined features F' are then processed separately for bounding box regression and class prediction.

The class prediction branch uses two standard convolution layers followed by a 1×1 convolution to output class probabilities. For the bounding box regression branch, we employ two Depthwise Separable Convolution (DSConv) layers. DSConv factorizes a standard convolution into a depthwise convolution (applying a single filter per input channel) followed by a pointwise convolution (1×1). This design significantly reduces the number of parameters and computational cost while efficiently processing fine-grained spatial features. The depthwise convolution preserves spatial information crucial for precise localization, and the pointwise convolution enables effective cross-channel feature fusion. This structure enhances the sensitivity to spatial details and positional relationships without a significant increase in computation, which is critical for accurate bounding box regression.

In summary, SDDetect leverages a multi-scale spatial attention mechanism to highlight regions of interest and utilizes depthwise separable convolutions to achieve a favorable balance between detection performance and computational efficiency.

3.4. NWD-WIoUv3: a composite loss for robust localization of miniature PCB defects

Accurate localization of miniature PCB defects remains a significant challenge for object detectors. The IoU-based loss functions commonly

used in models such as YOLOv8 present two major issues in such detection scenarios: first, for small targets, slight deviations in the predicted bounding box can lead to a sharp decline in IoU values, resulting in unstable gradient dynamics and impaired convergence. Second, when the predicted box and the ground truth box do not overlap, the IoU value becomes zero and fails to provide gradient information regarding their spatial relationship, causing the learning process to stall.

To overcome these limitations, we propose a novel loss function that synergistically combines the Normalized Wasserstein Distance (NWD) and Wise-IoU version 3 (WIoUv3). This design is predicated on their complementary properties for bounding box regression.

The WIoUv3 loss [30] introduces a non-monotonic focusing coefficient that dynamically moderates the gradient assignment from low-quality examples (outliers), thereby enhancing generalization and convergence stability. It is defined as:

$$L_{\text{WIoUv3}} = r L_{\text{WIoUv1}}, \quad (11)$$

where the dynamic focusing coefficient r is computed based on the degree of outlieriness β and a hyperparameter α :

$$r = \frac{\beta}{\delta \alpha^{\beta - \alpha}}, \quad \beta = \frac{L^{\text{IoU}}}{L^{\text{IoU}}}. \quad (12)$$

Here, L^{IoU} is the mean of the IoU loss, serving as a monotonic focusing factor, and $\overline{L^{\text{IoU}}}$ is a running mean that dynamically reflects the state of the training.

Conversely, the NWD metric [31] models bounding boxes as 2D Gaussian distributions and measures their similarity, offering superior performance for small objects. For a ground-truth box $R_{gt} = (c_x, c_y, w, h)$, it is modeled as a Gaussian distribution $\mathcal{N}(\mu_{gt}, \Sigma_{gt})$, where the mean μ_{gt} is the center coordinate (c_x, c_y) , and the covariance matrix Σ_{gt} is diagonal, derived from the width and height $(\frac{w^2}{4}, 0; 0, \frac{h^2}{4})$. The Wasserstein distance W_2^2 between two Gaussian distributions $\mathcal{N}_p$ (prediction) and $\mathcal{N}_{gt}$ (ground truth) is calculated as:

$$W_2^2(\mathcal{N}_p, \mathcal{N}_{gt}) = \|\mu_p - \mu_{gt}\|_2^2 + \text{Tr} \left(\Sigma_p + \Sigma_{gt} - 2(\Sigma_p^{1/2} \Sigma_{gt} \Sigma_p^{1/2})^{1/2} \right). \quad (13)$$

or the axis-aligned Gaussian distributions used in NWD, this simplifies to a computationally efficient form. The NWD is then obtained by

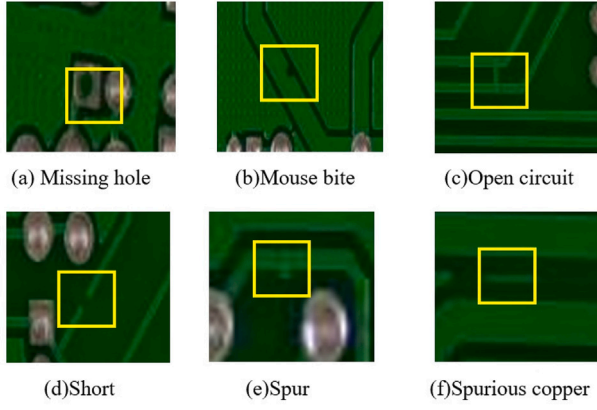


Fig. 6. PCB defect types in the PKU-Market-PCB datasets.

Table 2

Data augmentation techniques for PCB defect detection.

Augmentation methods	Methods description
Add noise cutout	Add Gaussian noise, with intensity in the range of [0, 255]. One occlusion area is applied, with side length in the range [50, 100] pixels. Monte Carlo rejection sampling ensures that the IoU between the occlusion area and the target box does not exceed 0.5.
Mirror	Random horizontal/vertical/diagonal mirroring is applied, with each type having a probability of 33 %.
Brightness adjustment shift	The brightness adjustment range is from 35 % to 99 %. Random shifting is applied in horizontal, vertical, or both directions, with the shift limited to one-third of the maximum distance from the bounding box to the image borders, ensuring the box stays within the image.
Rotation	The rotation angle range is $[-5^\circ, +5^\circ]$.

normalizing this distance to a [0, 1] range:

$$\text{NWD} = \exp \left(- \frac{\sqrt{W_2^2(\mathcal{N}_p, \mathcal{N}_{gt})}}{C} \right), \quad (14)$$

where C is a constant scaling factor. The associated loss is $\text{LNWD} = 1 - \text{NWD}$.

While WIoUv3 provides robust learning signals for general cases, it remains somewhat susceptible to the precise localization of minuscule targets. NWD excels in this regime but may offer less precise overlap

optimization for larger objects. To harness the strengths of both, we formulate a composite loss function:

$$L_{\text{NWD-WIoUv3}} = \gamma \cdot L_{\text{NWD}} + (1 - \gamma) \cdot L_{\text{WIoUv3}} \quad (15)$$

where γ is a balancing hyperparameter meticulously tuned for the PCB defect detection task (see Section 5.3 for the ablation study). This fusion enables the model to leverage the spatial modeling prowess of NWD for initial small defect capture and the refined convergence properties of WIoUv3 for precise box regression, culminating in enhanced accuracy and robustness across diverse defect sizes and complex backgrounds.

4. Experimental setup and evaluation criteria

This section describes the features of the PCBs datasets utilized in this research, as well as the experimental setup and the evaluation criteria used to assess the studied model's performance.

4.1. Structural characteristics and augmentation methods of the public datasets

This study utilizes the publicly available PCB defect dataset provided by the Intelligent Robotics Open Laboratory at Peking University (PKU-Market-PCB, <https://robotics.pkusz.edu.cn>). As shown in Fig. 6, the defect types include Missing hole, Open circuit, Mouse bite, Short (short circuit), Spurious (spurious copper), and Spur. The dataset contains only 1386 images.

The PKU-Market-PCB dataset is limited in size, so data augmentation techniques were applied to simulate industrial challenges. These include noise addition, Cutout, Mirroring, Brightness Adjustment, Shifting, and Rotation, replicating factors like uneven lighting and electromagnetic interference. The methods are detailed in Table 2 and Fig. 7. Each image also underwent compound augmentation using two or more methods.

After data augmentation, the dataset expanded to 7621 images. The augmentation distribution for each defect category is shown in Fig. 9. Fig. 8 presents the total label count for each defect type, the distribution of defect positions, and the relative positions of the true box centers in the image. The dataset is split into training, validation, and test sets with a 7:2:1 ratio.

4.2. Experimental environment and parameter settings

The experimental environment is summarized in Table 3. The experimental parameter settings are shown in Table 4. All unspecified hyper parameters followed the default settings of the YOLOv8 framework [32]. During the experiment, we conducted 5 repeated trials to assess the statistical significance of the studied model's results using p-value testing.

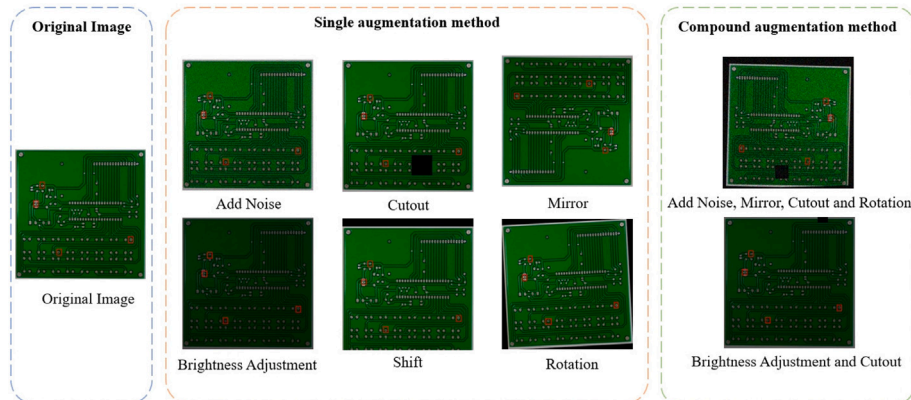


Fig. 7. Examples of data augmentation techniques.

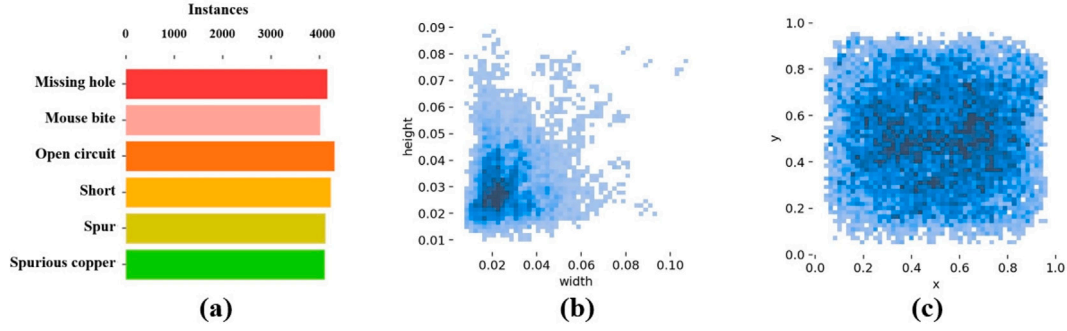


Fig. 8. (a) The total label count for each defect type; (b) The relative size of each true box in the image; (c) The relative position of each true box center in the image.

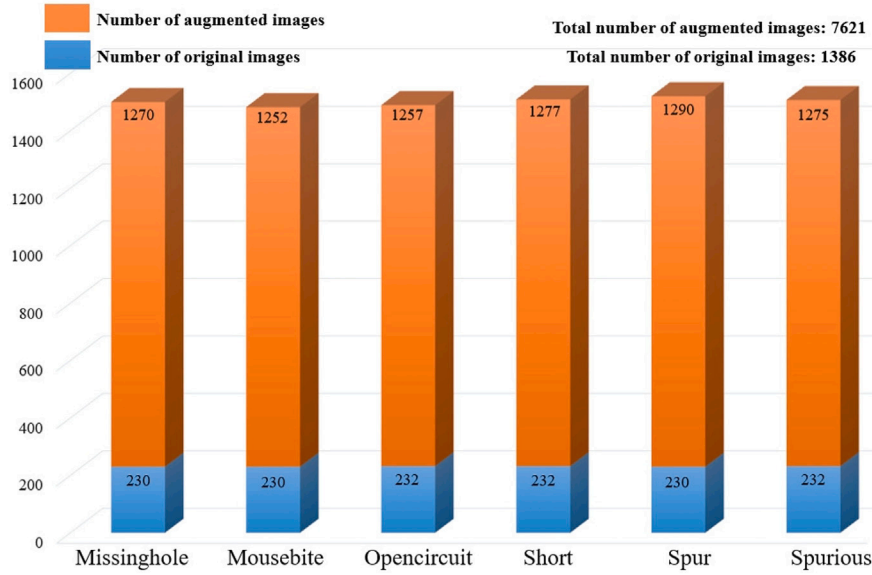


Fig. 9. Detailed PCB defect categories in PKU-Market-PCB.

Table 3
Experimental configuration.

Configuration	Parameter
GPU	Nvidia GeForce RTX 4060 Ti
CPU	12th Gen Intel(R) Core(TM) i5-12400 F
Operating System	Windows 11 (Enterprise Edition)
Accelerated environment	CUDA 11.8
Libraries	PyTorch-GPU 2.0.1
Development environment	PyCharm 2024

Table 4
Input parameter settings for numerical experiments.

Parameters	Value
Image_size	640 × 640
Learning_Rate	1e−2
optimizer	SGD
Weight_decay	5e−4
Batch_size	16
Workers	8
epoch	300
early stopping	50

4.3. Statistical significance testing

The P-value is a measure of the statistical significance of the results of a hypothesis test. It represents the probability of observing the current experimental data, or more extreme data, under the assumption that

the null hypothesis (i.e., there is no significant difference between the two groups) is true. The smaller the P-value, the less likely it is to observe such a difference under the null hypothesis, indicating that the experimental results are more statistically significant.

Hypothesis Testing: *Null Hypothesis:* We assume that there is no significant performance difference between model A and model B.

Alternative Hypothesis: We assume that there is a significant performance difference between model A and model B.

P-value Calculation: If the P-value is smaller than the predetermined significance level (0.05), we reject the null hypothesis, indicating that the difference between the two models is significant. If the P-value is greater than 0.05, we fail to reject the null hypothesis, indicating that the difference is not significant.

The *t*-test is a statistical method commonly used to compare whether there is a significant difference between the means of two groups. In this paper, we use the independent samples *t*-test to compare the performance differences (e.g., mAP, precision, recall, etc.) between the proposed YOLO-DTS model and the baseline model (e.g., YOLOv8).

The *t*-test formula is given by:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{X}_1, \bar{X}_2$ are the means of model 1 and model 2, respectively, s_1^2, s_2^2 are the variances of model 1 and model 2, respectively, and n_1, n_2 are the sample sizes of model 1 and model 2, respectively.

To perform the t -test, we first calculate the means and variances of the performance metrics (e.g., precision, recall, etc.) for both models based on different experiments. Then, we use the above formula to compute the t -value. Finally, we compare the calculated t -value with the critical value from the t -distribution table to determine whether the difference is significant.

4.4. Performance metrics

The evaluation metrics used in this study include Precision (P), Recall (R), F1-score ($F1$), Mean Average Precision (mAP) at IoU thresholds of 0.50 (mAP_{50}) and 0.95 (mAP_{95}), the number of model *Parameters*, Giga Floating Point Operations per Second ($GFLOPs$), and Frames Per Second (FPS).

True Positives (TP) refer to the number of correctly detected target instances, while False Positives (FP) represent the number of incorrectly detected targets. True Negatives (TN) denote the number of correctly identified non-target instances, and False Negatives (FN) refer to the missed detections. The total number of samples is given by the sum $TP + FP + TN + FN$. Based on these, the Precision (P) and Recall (R) are computed using Eqs. (16) and (17), respectively.

$$P = \frac{TP}{TP + FP} \quad (16)$$

$$R = \frac{TP}{TP + FN} \quad (17)$$

To balance the comprehensive metric of precision and recall, we use the F1-score as the harmonic mean, as shown in the Eq. (18). This metric is used in cases of class imbalance and provides a more comprehensive evaluation of YOLO-DTS model's performance.

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (18)$$

Average Precision (AP) is defined as the area under the *Precision – Recall* curve, as demonstrated in Eq. (19). Meanwhile, mean Average Precision (mAP) represents the average precision across all categories (k) in Eq. (20).

$$mAP = \frac{\sum_{i=1}^k AP_i}{k} \quad (19)$$

$$AP = \int_0^1 P(R) dR \quad (20)$$

In Eqs. (19) and (20), $P(R)$ represents precision as a function of recall, and k is the category number. Since Intersection over Union (IoU)

evaluates prediction accuracy, mAP can be calculated at various IoU thresholds. mAP_{50} is the mean precision at an IoU threshold of 0.5, while mAP_{95} averages precision across IoU thresholds from 0.5 to 0.95. $GFLOPs$ represents the floating-point operations required for a single forward pass, reflecting computational intensity. *Parameters* denote the total trainable parameters, indicating learning capacity. Together, these metrics assess the model's scale and complexity. FPS is a crucial metric for evaluating the performance of image and video processing systems, as it quantifies the number of frames processed or displayed per second.

5. Model performance and comparative analysis

This section presents experimental results, including comparisons with baselines, ablation studies, and generalization tests, demonstrating the effectiveness and robustness of YOLO-DTS.

5.1. Training results of YOLO-DTS

Our method employed an early stopping mechanism set at 50 epochs during training on the enhanced PKU-Market-PCB dataset, but the stopping condition was never triggered throughout the full 300 epochs of training. As shown in Fig. 10 (a), all smoothed loss curves (smooth) demonstrated a steady decline until around epoch 200, after which they stabilized, indicating full convergence in the later stages. Minor fluctuations in the raw data curves (result) are considered normal. At epoch 300, the final training and validation loss values were closely aligned with consistent trends, suggesting no significant overfitting occurred. Therefore, setting the total training epochs to 300 was reasonable and sufficient.

The calculation of the confidence level is typically based on the distribution of sample means. In our experiment, we performed 5 independent trials and calculated the mean and standard deviation of the results. Based on these data, we use the following formula to calculate the 95 % confidence interval:

$$ConfidenceInterval = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \quad (21)$$

Where, the $\bar{X}$ is the mean of the experimental results, $Z_{\alpha/2}$ is the critical value from the standard normal distribution, typically $Z_{\alpha/2} = 1.96$ for a 95 % confidence interval, σ is the standard deviation of the experimental results, and n is the sample size (number of independent trials).

The F1 score-confidence curve is shown in Fig. 10 (b), where the F1 score reaches its highest value of 0.78 (± 0.1) at a confidence level

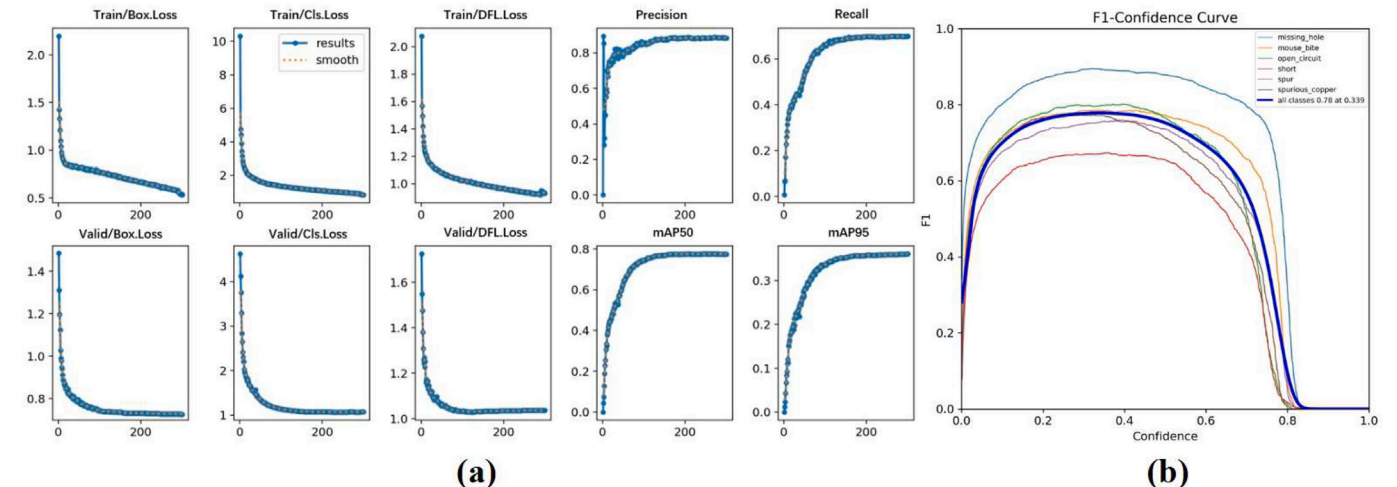


Fig. 10. (a) Training results of the YOLO-DTS Model. (b) F1-confidence curve of the YOLO-DTS model.

Table 5

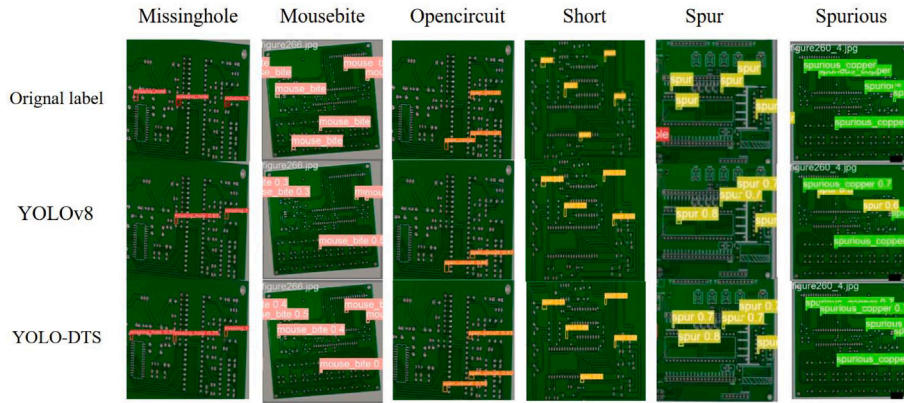
The performance comparison between YOLO-DTS and YOLOv8 in terms of metrics.

Dataset	Model	P (%)	R (%)	$F1$	mAP_{50} (%)	mAP_{95} (%)	Parameter(M)	$GFLOPs$	FPS
PKU-Market-PCB	YOLOv8(baseline)	87.2 ± 0.1	61.9 ± 0.1	72.4	74.6 ± 0.1	34.3 ± 0.1	3.0	8.1	474.89
	YOLO-DTS (ours)	89.5 ± 0.1	73.0 ± 0.2	80.4	82.4 ± 0.1	38.5 ± 0.1	2.7	5.7	420.61
Augmentation PKU-Market-PCB	YOLOv8(baseline)	83.4 ± 0.1	62.4 ± 0.1	71.1	70.0 ± 0.1	32.5 ± 0.1	3.0	8.1	367.15
	YOLO-DTS (ours)	88.4 ± 0.1	69.9 ± 0.2	78.0	77.5 ± 0.1	36.1 ± 0.1	2.7	5.7	336.40

Table 6

The performance comparison between YOLO-DTS and YOLOv8 in terms of precision.

Dataset	Model	Missinghole	Mousebite	Opencircuit	Short	Spur	Spurious
PKU-Market-PCB	YOLOv8(baseline)	91.8 ± 0.1	91.1 ± 0.1	89.7 ± 0.2	86.6	80.4 ± 0.1	83.6 ± 0.1
	YOLO-DTS (ours)	96.2 ± 0.2	88.2 ± 0.1	95.1 ± 0.1	84.3 ± 0.1	89.7 ± 0.1	83.8 ± 0.1
Augmentation PKU-Market-PCB	YOLOv8(baseline)	83.3 ± 0.1	83.0 ± 0.1	84.8 ± 0.1	79.8 ± 0.1	79.6 ± 0.1	84.0 ± 0.1
	YOLO-DTS (ours)	92.1 ± 0.1	86.5 ± 0.1	89.7 ± 0.1	82.0 ± 0.1	87.7 ± 0.1	91.8 ± 0.1

**Fig. 11.** Detection performance of YOLO-DTS vs YOLOv8 for PCB defects on PKU-Market-PCB.

of $0.339 (\pm 0.02)$ ($p < 0.01$). As the confidence increases, its F1 score gradually rises and stabilizes, indicating that at higher confidence levels, our model is able to make more accurate predictions with a good balance between precision and recall.

5.2. Comparison with baseline

To evaluate the efficacy of data augmentation in enhancing model robustness, experiments were conducted on both the original and augmented versions of the PKU-Market-PCB dataset. The augmented set incorporates more complex geometric transformations, illumination variations, and simulated occlusions, significantly increasing data diversity and complexity.

As shown in Tables 5 and 6, the augmented dataset posed greater challenges for model training. Overall performance metrics slightly declined on the augmented data. For example, YOLO-DTS achieved an mAP_{50} of $77.5 \% \pm 0.1 \%$ ($p < 0.01$) on the augmented set, compared to $82.482.4 \% \pm 0.1 \%$ ($p < 0.01$) on the original data. An exception was the Spurious defect category, where accuracy improved. This gain is attributed to the augmentation strategy effectively simulating diverse morphological and textural features of Spurious defects, which helped reduce confusion between true defects and complex background noise. For all other defect types, accuracy decreased, consistent with the expectation that the more complex data distribution mitigates overfitting and improves generalization.

As shown in Table 5, YOLO-DTS consistently outperforms the YOLOv8 baseline across all key metrics on both datasets. On the augmented set, it achieves an mAP_{50} of $77.5 \% \pm 0.1 \%$ ($p < 0.01$), a 7.5 % absolute improvement over the baseline, confirming the efficacy of its architectural design. Furthermore, as detailed in Table 6, YOLO-DTS

attains the highest accuracy in five out of the six PCB defect categories, demonstrating particularly strong performance on critical defects such as “Missinghole” and “Opencircuit”.

YOLO-DTS also exhibits high model efficiency, with only 2.7 M parameters and 5.7 GFLOPs—representing reductions of 13.3 % in parameters and 29.6 % in computations compared to YOLOv8. Although FPS is slightly lower, the decrease is marginal relative to the substantial gains in accuracy. In industrial inspection applications, where detection reliability is paramount, this trade-off is highly advantageous, as higher accuracy significantly reduces the risk of misdetections and production line disruptions.

As shown in Figs. 11 and 12, our method more effectively reduces both false positives and false negatives across all defect types compared to YOLOv8. However, in complex environments, certain defects remain challenging to accurately identify, primarily due to the following reasons: First, extreme lighting conditions, metallic reflections, or shadow interference can obscure the visual characteristics of defects, leading to a lack of apparent information. Second, some defects exhibit highly irregular shapes or closely resemble normal structural textures, resulting in low feature discriminability. Third, small-scale defects or incomplete regions have extremely low resolution in the image, making it difficult to provide sufficient discriminative responses.

To investigate how YOLO-DTS achieves higher performance, we used Grad-CAM [33] to visualize attention heatmaps during defect detection. As shown in Fig. 13, YOLO-DTS exhibits more focused activation on actual defects compared to YOLOv8, which often shows scattered attention and background over-activation. For example, when detecting Spur defects, YOLO-DTS accurately highlights the full defect structure with clear boundaries, while YOLOv8’s responses spread to irrelevant background regions. Similarly, for geometric defects like Missinghole,

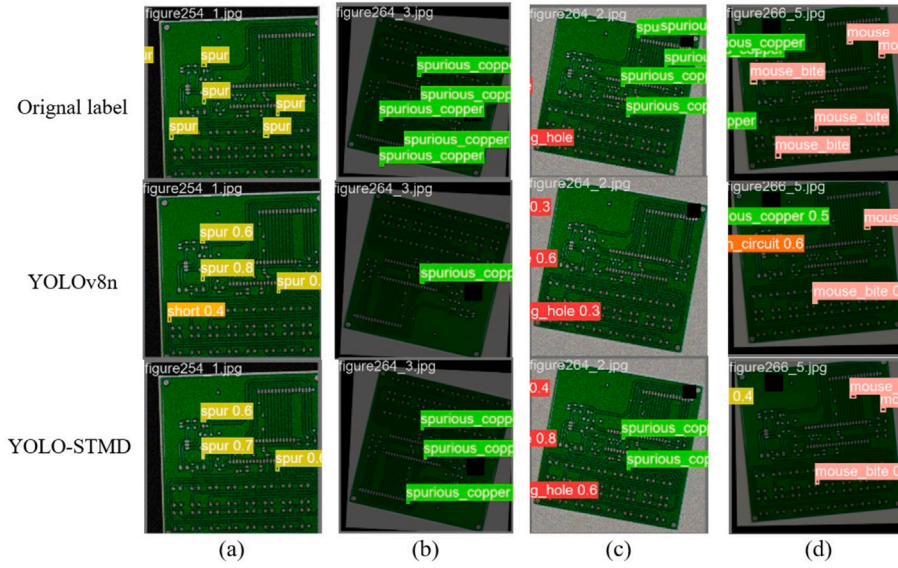


Fig. 12. Detection performance of YOLO-DTS vs YOLOv8 for PCB defects on augmentation PKU-Market-PCB.

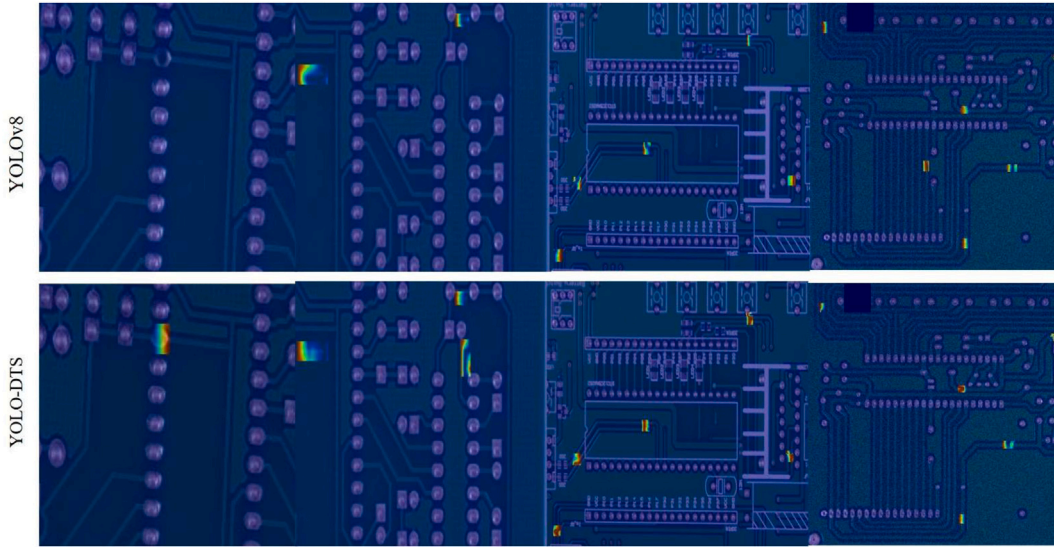


Fig. 13. Heatmap of YOLO-DTS vs YOLOv8 for PCB defects.

YOLO-DTS concentrates precisely on the missing area. These results confirm that YOLO-DTS learns more discriminative features and effectively suppresses background interference.

Through the confusion matrix, we can intuitively identify the false negatives and false positives that occur in the model's detection. As shown in Fig. 14, false positives between Spur and SpuriousCopper defects are observed. Spur defects typically manifest as protrusions or serrated edges on the circuit board. These defects are small and usually occur at the edges or gaps of the board, and their shape shares similarities with copper traces. On the other hand, SpuriousCopper defects refer to the presence of excessive copper layers or irregular copper pieces on the circuit board. These irregular copper pieces are often found near other copper traces or the edges of the board. Since both defects involve abnormalities in the copper layers on the circuit board and share small, similar shapes, the model tends to confuse them with each other.

5.3. The impact of different loss function weights on the YOLO-DTS model

To investigate the impact of different weights of NWD and WIoUv3 in the loss function on the YOLO-DTS model's performance, we conducted

experiments with varying values of γ . The experimental results are shown in Table 7.

When $\gamma = 0$, the loss function relies entirely on WIoUv3, which is sensitive to small positional offsets of the target, resulting in a precision of only 87.1 %. When $\gamma = 1$, the loss function is based solely on NWD, which struggles with target overlaps and fails to capture spatial layouts and geometric differences, leading to a precision of just 86.5 %. It can be seen that when NWD's contribution is higher than WIoUv3's ($\gamma > 0.5$), while defects are localized precisely, classification accuracy suffers, lowering detection efficiency compared to when WIoUv3's contribution is greater than NWD's ($\gamma < 0.5$). However, if WIoUv3's contribution is too large, it fails to address small target position offsets effectively, potentially leading to overfitting on detailed features. The experimental results show that when $\gamma = 0.35$, with 35 % NWD and 65 % WIoUv3, the loss function achieves the best performance, with a precision of 88.4 %, outperforming other comparative experiments.

To evaluate the advantages of our proposed loss function for PCB defect detection, we conducted comparative experiments with state-of-the-art bounding box regression loss functions including EIoU [34], SIoU

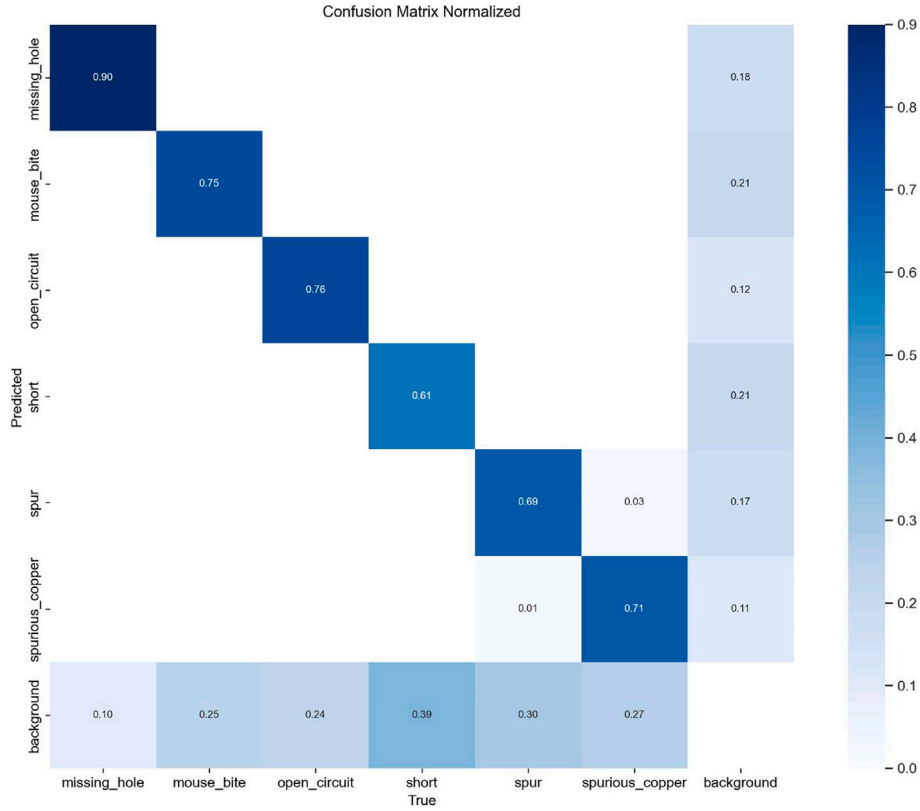


Fig. 14. Normalized confusion matrix of the YOLO-DTS model.

Table 7

The impact of different weights on our method.

Value of γ	P (%)	R (%)	F1	mAP_{50} (%)	mAP_{95} (%)
0(WIoUv3)	87.1 ± 0.1	64.2 ± 0.1	73.9	76.7 ± 0.1	35.6 ± 0.1
0.1	86.9 ± 0.2	69.0 ± 0.1	76.9	76.5 ± 0.2	34.9 ± 0.1
0.25	87.7 ± 0.2	69.4 ± 0.2	77.4	77.1 ± 0.2	35.4 ± 0.2
0.35	88.4 ± 0.1	69.9 ± 0.1	78.0	77.5 ± 0.1	36.1 ± 0.1
0.4	87.6 ± 0.1	69.1 ± 0.1	77.2	77.3 ± 0.1	35.5 ± 0.1
0.45	87.6 ± 0.2	68.1 ± 0.1	76.6	76.3 ± 0.1	35.4 ± 0.1
0.5	87.3 ± 0.1	69.5 ± 0.2	77.3	76.9 ± 0.1	35.5 ± 0.1
0.75	87.4 ± 0.1	68.4 ± 0.1	76.7	76.3 ± 0.1	35.6 ± 0.1
1.0(NWD)	86.5 ± 0.2	69.0 ± 0.2	76.7	76.6 ± 0.2	35.3 ± 0.1

[35], and DIoU [36]. As shown in Table 8, EIoU achieved the lowest mAP_{50} of 74.4 % among all compared methods due to its strong reliance on IoU metric, making it particularly unsuitable for small object detection. DIoU attained a reasonable recall rate of 67.7 % but showed limitations in handling extreme scale variations, resulting in relatively low precision of 86.4 %. SIoU achieved balanced performance with an F1-score of 76.3 and mAP_{50} of 75.7 %, however, its recall rate remained at 67.5 % due to interference from complex backgrounds. Our proposed method achieved a precision of 88.4 %, which is 3.3 % higher than the original YOLOv8 loss function CIoU, and an mAP_{50} of 77.5 %, surpassing CIoU by 4.7 %. It also outperforms other methods, significantly enhancing the overall performance of PCB defect detection.

5.4. Ablation analysis of c2f_Deformable attention

This chapter presents an ablation study analyzing the impact of the scaling factor s and the use of residual connections in the C2f_DAT module on model performance.

The scaling factor s is used to control the range of the learnable offset (Δp), ensuring that the magnitude of the offset remains constrained.

Table 8

Comparison of different loss functions.

Loss function	P (%)	R (%)	F1	mAP_{50} (%)	mAP_{95} (%)
CIoU	85.1	64.5	73.3	74.1	33.4
EIoU	85.7	66.6	74.9	74.4	34.3
SIoU	87.8	67.5	76.3	75.7	35.4
DIoU	86.4	67.7	75.9	75.7	34.8
Ours	88.4	69.9	78.0	77.5	36.1

This helps maintain training stability and regulates the breadth of the model's receptive field. As shown in Table 9, the model achieves optimal performance across all evaluation metrics when $s = 2$. Further increasing s leads to a noticeable decline in all metrics, indicating that a larger offset range adversely affects performance.

PCB defects typically manifest as minor morphological anomalies within localized regions. These critical features are often densely distributed and located in close proximity to the defect center. A smaller offset range is sufficient to cover these relevant areas, allowing the model to accurately focus on these subtle anomalies. Moreover, PCB layouts are highly structured and regular, with relatively controllable background noise. An excessively large offset range may cause the attention mechanism to include irrelevant background areas far from the region of interest, thereby introducing distracting information and degrading recognition performance.

We compared the impact of incorporating residual connections in the C2f_DAT module on model performance, as shown in the Table 10. The experimental results indicate that the model with residual connections outperforms the one without in all evaluation metrics. Specifically, the model with residual connections achieves a precision of 88.4 %, which is 2.9 % higher than the model without residual connections. Residual connections facilitate the direct propagation of information across network

Table 9
Impact of scaling factor s on model performance.

	P (%)	R (%)	F1	mAP_{50} (%)	mAP_{95} (%)
$s = 2$	88.4	69.9	78.0	77.5	36.1
$s = 4$	86.9	69.0	76.9	76.5	34.9
$s = 6$	86.3	68.7	76.5	76.0	34.8
$s = 8$	86.4	68.6	76.4	76.3	35.1

Table 10
Effects of using residual connections on YOLO-DTS.

Residual connections	P (%)	R (%)	F1	mAP_{50} (%)	mAP_{95} (%)
✓	88.4	69.9	78.0	77.5	36.1
	85.5	67	75.1	75.3	33.8

layers, mitigating the risk of information loss in deep networks. This is particularly beneficial for fine-grained tasks such as PCB defect detection, where residual connections help preserve more detailed features, enabling the model to better identify subtle defects.

5.5. The impact of different dropout rates on the DTdown

As illustrated in Fig. 15, as the dropout rate increases, its influence on DTdown shows an approximately linear decline. This trend is likely due to excessive neuron dropout, which discards critical feature representations and leads to underfitting. Distinct performance gaps appear among different dropout rates after approximately 50 epochs. Notably, at a dropout rate of 0.4, our model prematurely converges by the 248th

epoch, achieving only 81.8 % precision—6.6 % lower than the 88.4 % achieved with our selected rate of 0.1. These results show that a lower dropout rate (e.g., 0.1) effectively mitigates overfitting while maintaining the model's ability to learn discriminative features. Therefore, we adopt a dropout rate of 0.1 for the DTdown module.

5.6. Ablation study on YOLO-DTS components

We conducted ablation experiments on the three proposed modules to evaluate their impact on model performance, with all hyper parameters kept consistent throughout the experiments. The results are shown in Table 11.

When each of the three modules is added individually, our model's performance improves noticeably. Specifically, DTdown reduces the amount of data processed by convolution through different branches and makes the network more compact by using both average pooling and max pooling. This results in a precision of 87.8 %, a 4.3 % improvement, while reducing the parameter count by 0.5 M. Adding C2f-Deformable Attention does not significantly alter the overall structure of YOLO-DTS, maintaining almost the same number of parameters and $GFLOPs$, while achieving a 1.9 % precision increase. SDDetect, by using depth-wise separable convolutions, reduces the parameter count the most, by up to 0.6 M, while achieving a 2.6 % precision increase.

However, when both DTdown and C2f-Deformable Attention are combined, accuracy improves to 87.5 %, but the parameter count increases to 3.2 M, adding a significant computational burden. When SDDetect and DTdown are combined, the parameter count is 2.6 M, close to YOLO-DTS, but mAP_{50} drops to 34.4 %, which is 1.7 % lower than YOLO-DTS. When C2f-Deformable Attention and SDDetect are

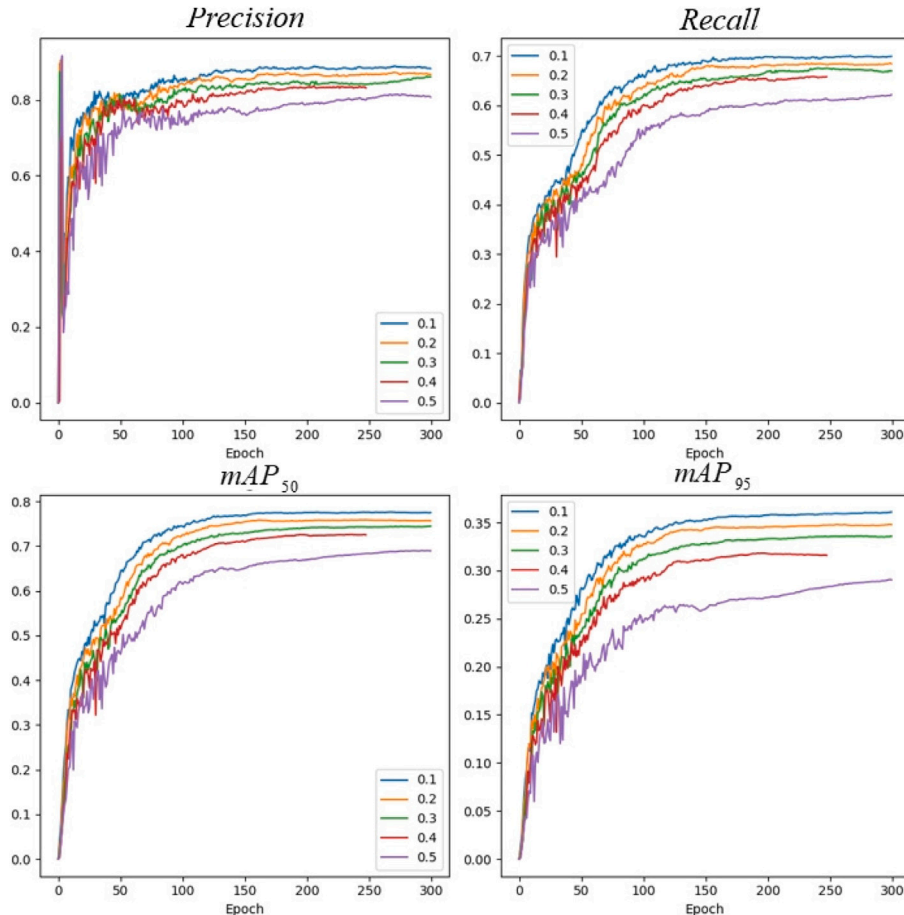


Fig. 15. Effect of dropout rates on precision and mAP_{50} .

Table 11
Ablation experiment results.

Index	DTdown	C2f-Deformable attention	SDDetect	P(%)	R(%)	F1	mAP_{50} (%)	mAP_{95} (%)	Parameters(M)	GFLOPs(G)
1	–	–	–	83.5	62.4	76.8	71.0	32.5	3.0	8.1
2	✓	–	–	87.8	68.3	76.8	75.8	34.9	2.5	5.5
3	–	✓	–	85.4	63.5	72.8	72.1	33.3	3.0	8.1
4	–	–	✓	86.1	68.1	76.0	75.6	35.0	2.4	5.8
5	✓	✓	–	87.5	68.7	76.9	76.3	34.5	3.2	8.1
6	✓	–	✓	87.2	68.2	76.2	76.5	34.4	2.6	5.7
7	–	✓	✓	86.6	61.8	76.5	75.3	35.1	2.4	5.8
8 (ours)	✓	✓	✓	88.4	69.9	78.0	77.5	36.1	2.7	5.7

Table 12
Performance metrics of various models on the augmented PKU-Market-PCB datasets.

Model	P (%)	R (%)	F1	mAP_{50} (%)	mAP_{95} (%)	Parameter(M)	GFLOPs	FPS
Faster-RCNN [37]	47.9 ± 0.3	40.9 ± 0.2	44.1	41.1 ± 0.3	29.9 ± 0.3	41.7	89.4	17.6
RT-DETR-H [38]	72.4 ± 0.2	38.8 ± 0.1	50.6	46.3 ± 0.1	36.1 ± 0.1	284.5	100.6	75.2
YOLOv6 [44]	83.7 ± 0.2	64.8 ± 0.1	73.0	72.9 ± 0.2	33.5 ± 0.1	4.2	11.8	554.3
YOLOv8(baseline)	83.4 ± 0.1	62.4 ± 0.1	71.1	70.0 ± 0.1	32.5 ± 0.1	3.0	8.1	367.1
YOLOv8s [32]	86.7 ± 0.2	67.5 ± 0.1	75.9	75.4 ± 0.1	35.4 ± 0.1	11.1	28.4	297.5
Gold-YOLO [42]	84.1 ± 0.2	62.9 ± 0.2	71.9	70.5 ± 0.1	32.3 ± 0.2	5.9	10.2	208.5
YOLOv9 [26]	83.4 ± 0.1	60.8 ± 0.1	70.4	67.9 ± 0.1	32.2 ± 0.1	2.0	7.9	154.2
YOLOv10 [39]	84.3 ± 0.3	65.8 ± 0.1	73.9	74.4 ± 0.1	35.7 ± 0.2	2.6	8.2	115.2
YOLOv11 [40]	84.8 ± 0.2	63.8 ± 0.3	73.5	72.1 ± 0.2	33.7 ± 0.2	2.5	6.3	138.9
YOLOv12 [41]	84.8 ± 0.1	47.5 ± 0.1	60.8	58.7 ± 0.1	27.6 ± 0.1	2.5	6.3	89.4
YOLO-SSW [7]	88.9 ± 0.1	70.5 ± 0.1	78.6	79.0 ± 0.1	37.2 ± 0.1	4.3	11.2	171.9
YOLO-RRL [43]	85.9 ± 0.1	66.8 ± 0.2	75.1	74.6 ± 0.1	34.2 ± 0.1	2.4	5.8	308.4
Literature [23]	82.2 ± 0.2	60.5 ± 0.1	69.7	67.9 ± 0.1	30.8 ± 0.2	1.7	5.1	372.3
YOLO-DTS (ours)	88.4 ± 0.1	69.9 ± 0.2	78.0	77.5 ± 0.1	36.1 ± 0.1	2.7	5.7	336.4

combined, the parameter count reduces to 2.4 M, 0.2 M lower than YOLO-DTS, but mAP_{50} falls to 75.3 %, which is 2.2 % lower than YOLO-DTS. Overall, combining all three modules fully leverages their performance advantages, achieving the best balance between efficiency and accuracy within the constraints of limited resources.

5.7. Performance comparison between compared models

To demonstrate the advantages of the proposed method, we compare YOLO-DTS with current mainstream object detection models, including the representative two-stage model Faster R-CNN [37], the region-based transformer for end-to-end object detection RT-DETR-H [38], as well as popular YOLO-series models such as YOLOv8 small (YOLOv8s) [32], YOLOv9 [26], YOLOv10 [39], YOLOv11 [40], YOLOv12 [41], Gold-YOLO [42], and recent advances in PCB defect detection including Literature [23], YOLO-RRL [43], and YOLO-SSW [7]. The models are evaluated using multiple metrics to assess their performance in PCB defect detection tasks. Detailed comparative results are provided in Tables 12 and 13.

The representative two-stage detector, Faster R-CNN, suffers from a large parameter size and slow inference speed due to its complex network architecture, resulting in relatively low precision and recall which fail to meet the high real-time requirements of industrial quality inspection. Although the Transformer-based RT-DETR-H model achieves a precision of 72.4 %, its recall is only 38.8 %, and its parameter count reaches 284.5 M, leading to significantly high deployment costs.

The YOLO series has gained research attention for its favorable balance between speed and accuracy, yet notable performance variations exist among its variants. YOLOv8s delivers excellent overall detection accuracy, but with 11.1 M parameters and 28.4 GFLOPs computational cost, it imposes the heaviest burden within the YOLO family, making it less suitable for extremely lightweight scenarios. YOLOv9 reduces the parameter count to 2.0 M through programmable gradient information and a generalized efficient layer aggregation network; however, it achieves only 60.8 % recall on this dataset, indicating limited capability in detecting small defects.

YOLOv10 adopts an NMS-free design and consistent matching strategy, achieving 74.4 % mAP_{50} with only 2.6 M parameters. Nevertheless, there remains room for improvement in recognizing certain categories such as Mousebite. YOLOv11 attains 72.1 % mAP_{50} with 2.5 M parameters via optimized training strategies and architectural design, but its performance on complex defects like Spurious is still not prominent. YOLOv12 maintains a parameter count of 2.5 M, yet its recall drops significantly to 47.5 %, suggesting a structural design poorly matched to PCB defect detection tasks and resulting in severe under-detection issues.

The model proposed by Yu et al. [23] integrates lightweight techniques such as GhostNet and depthwise separable convolutions into a PCB defect detection framework. With only 1.7 million parameters and a computational cost of 5.1 GFLOPs, the model achieves an inference speed of 372.3 FPS. However, its mAP_{50} is 67.9 %, which is nearly 10 % ($p < 0.01$) lower than the approach presented in this study. YOLO-RRL, designed specifically for PCB defect detection, has 2.4 million parameters and an inference speed of 308.4 FPS, but its accuracy is limited to 85.9 %. The YOLO-SSW model improves small target detection performance by incorporating a P2 feature layer and an enhanced loss function, reaching an mAP_{50} of 79.0 %. However, this model incurs a high computational cost of 11.2 GFLOPs.

As shown in Table 12, while YOLO-DTS ranks second in comprehensive metrics like mAP_{50} and F1-score, its core strength lies in its superior performance-efficiency trade-off. The top-performing model, YOLO-SSW, requires 59 % more parameters and 96 % higher computational cost than YOLO-DTS. In contrast, YOLO-DTS achieves competitive performance with one of the lowest resource footprints (2.7 M parameters, 5.7 GFLOPs) while maintaining a high inference speed (336.4 FPS). This demonstrates that YOLO-DTS is a more efficient and practical solution for industrial deployment.

The proposed method exhibits differentiated performance across various defect detection tasks. For Missinghole defects, our method achieves an accuracy of 92.1 %, slightly lower than YOLO-SSW's 93.4 % ($p < 0.01$). In Mousebite detection, the model achieves an accuracy of 86.5 %, outperforming all comparison models except for our model, which has

an accuracy of 87.4 %. For complex defects such as Opencircuit, the proposed model achieves the highest accuracy of 89.7 %. In detecting Short defects, the accuracy of YOLO-DTS is 82.0 %, slightly lower than YOLO-SSW's 87.3 % and YOLOv11's 83.0 % ($p < 0.01$). For Spur defects, our model achieves an accuracy of 87.7 %, second only to YOLO-SSW and YOLOv8s, but significantly outperforming other models. In the case of Spurious copper defects, which are prone to false positives, our method attains the highest detection accuracy at 91.8 %, demonstrating excellent robustness against interference.

These results indicate that the YOLO-DTS model exhibits strong generalization and discriminative ability, with good robustness across a

range of defect types. However, there is still room for improvement in the recognition of certain defect categories.

5.8. Generalization experiment

To assess the generalization ability of YOLO-DTS, we conducted experiments on the PCB Defect datasets (DeepPCB) and Aluminum Profiles Defect datasets (APSPC). The comprehensive results are summarized in Table 14.

On the DeepPCB datasets, our method demonstrated superior performance, achieving a mAP_{95} of $79 \% \pm 0.1 \%$, surpassing the baseline

Table 13

The performance comparison between YOLO-DTS and YOLOv8 in terms of precision.

Model	Missinghole	Mousebite	Opencircuit	Short	Spur	Spurious
Faster-RCNN [37]	69.1 ± 0.1	43.8 ± 0.2	38.2 ± 0.1	50.9 ± 0.1	37.7 ± 0.1	50.2 ± 0.2
RT-DETR-H [38]	73.3 ± 0.2	72.8 ± 0.3	71.7 ± 0.1	69.2 ± 0.1	70.0 ± 0.1	75.6 ± 0.2
YOLOv6 [44]	92.6 ± 0.1	86.6 ± 0.2	85.0 ± 0.2	82.5 ± 0.1	85.9 ± 0.1	84.1 ± 0.1
YOLOv8(baseline)	83.3 ± 0.1	83.0 ± 0.1	84.8 ± 0.1	79.8 ± 0.1	79.6 ± 0.1	84.0 ± 0.1
YOLOv8s [32]	91.6 ± 0.1	83.7 ± 0.1	87.3 ± 0.2	82.0 ± 0.2	88.3 ± 0.1	88.0 ± 0.1
Gold-YOLO [42]	90.6 ± 0.2	82.4 ± 0.1	84.2 ± 0.2	79.9 ± 0.1	84.4 ± 0.3	86.1 ± 0.1
YOLOv9 [26]	87.6 ± 0.1	82.0 ± 0.1	84.8 ± 0.1	80.6 ± 0.1	79.5 ± 0.1	84.6 ± 0.1
YOLOv10 [39]	88.6 ± 0.3	82.6 ± 0.2	86.9 ± 0.2	79.8 ± 0.3	82.2 ± 0.2	85.9 ± 0.3
YOLOv11 [40]	90.4 ± 0.2	83.5 ± 0.1	85.4 ± 0.1	83.0 ± 0.2	85.6 ± 0.2	85.5 ± 0.2
YOLOv12 [41]	90.1 ± 0.1	82.3 ± 0.1	85.1 ± 0.1	82.4 ± 0.1	84.0 ± 0.1	84.7 ± 0.1
YOLO-SSW [7]	93.4 ± 0.1	87.4 ± 0.1	87.7 ± 0.1	87.3 ± 0.1	87.2 ± 0.1	89.2 ± 0.1
YOLO-RRL [43]	91.0 ± 0.1	84.2 ± 0.1	84.9 ± 0.1	80.7 ± 0.1	86.4 ± 0.1	88.3 ± 0.1
Literature [23]	87.8 ± 0.2	82.6 ± 0.2	85.2 ± 0.1	78.9 ± 0.2	77.3 ± 0.1	81.7 ± 0.1
YOLO-DTS (ours)	92.1 ± 0.1	86.5 ± 0.1	89.7 ± 0.1	82.0 ± 0.1	87.7 ± 0.1	91.8 ± 0.1

Table 14

Experimental results on DeepPCB and APSPC datasets.

Model	DeepPCB			APSPC		
	P (%)	mAP_{50} (%)	mAP_{95} (%)	P (%)	mAP_{50} (%)	mAP_{95} (%)
Baseline	97.0 ± 0.2	98.4 ± 0.2	72.2 ± 0.2	69.3 ± 0.1	69.0 ± 0.1	40.8 ± 0.1
Our methods	98.0 ± 0.1	98.8 ± 0.1	79.0 ± 0.1	69.1 ± 0.1	71.2 ± 0.1	43.0 ± 0.1

Table 15

the comparison of mAP_{50} for all types of defects in DeepPCB.

Model	Background	Open	Short	Mouse bite	Spur	Copper
Baseline	96.9 ± 0.2	94.5 ± 0.1	97.3 ± 0.2	97.5 ± 0.3	99.3 ± 0.2	96.5 ± 0.1
Our methods	99.2 ± 0.1	95.4 ± 0.1	98.9 ± 0.1	97.7 ± 0.2	98.7 ± 0.1	98.3 ± 0.1

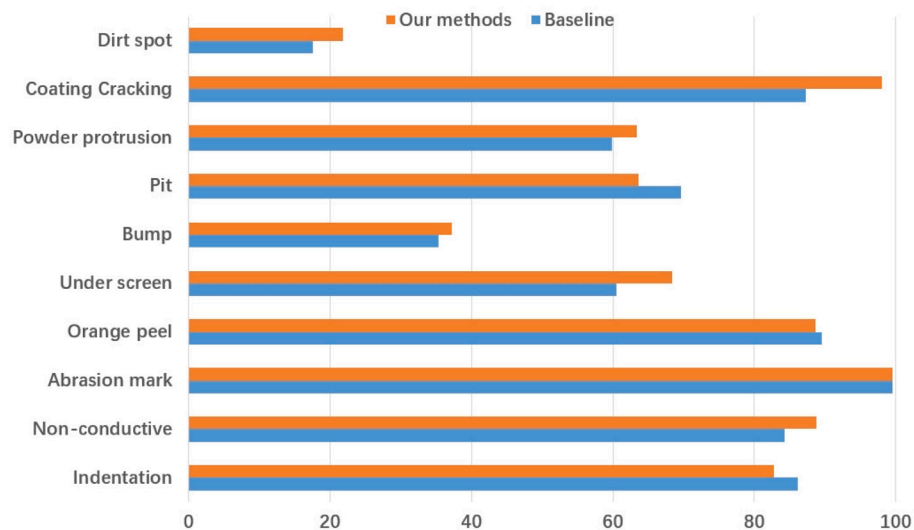


Fig. 16. Comparison of mAP_{50} for all type for APSPC datasets.

model's $72.2\% \pm 0.2\%$ by 6.8% ($p < 0.01$). The mAP_{50} precision for each defect category is provided in the Table 15, indicating that our method consistently outperforms the baseline across all defect types. This indicates that our method demonstrates good performance across most of the PCB datasets.

Aluminum profiles, a high-performance structural material based on aluminum, are widely used in industries such as electronics and aerospace manufacturing. Surface defects in aluminum profiles can present significant challenges for subsequent processing. In this study, we utilized a real-world dataset containing images of ten types of surface defects, including Indentation, Pit, and Non-conductive defects. Our proposed method achieved a precision of $69.3\% \pm 0.1\%$ on this dataset, slightly lower than the baseline model by 0.2% ($p < 0.01$). However, while the baseline model prioritized precision, it compromised recall, resulting in an mAP_{95} of only $40.8\% \pm 0.1\%$. In contrast, our method achieved an mAP_{95} of $43\% \pm 0.1\%$. As shown in Fig. 16, our approach outperformed the baseline model in detecting the majority of defect types in terms of mAP_{50} .

The experiments on the two industrial datasets demonstrate that our method exhibits strong adaptability and robustness across diverse datasets, highlighting its excellent generalization capabilities.

6. Conclusion

This paper presents a novel PCB defect detection model designed to excel in complex industrial environments. The proposed YOLO-DTS architecture introduces several key innovations: a dual-branch down-sampling module (DTdown) that integrates DualConv's grouping strategy with Transformer-based feature extraction to enhance defect detection capability while reducing computational overhead; the incorporation of Deformable Attention within the C2f module for effective handling of irregularly shaped defects; and the SDDetect detection head, which employs an augmented spatial attention mechanism, residual connections, and depthwise separable convolutions (DSConv) to improve localization and classification accuracy. Furthermore, a composite loss function combining Normalized Wasserstein Distance (NWD) and Wise-IoU v3 (WIoUv3) boosts the model's generalization performance.

To simulate challenging real-world conditions, we augmented the publicly available PKU-Market-PCB dataset. On this enhanced dataset, YOLO-DTS demonstrates superior detection performance, achieving 88.4% precision, 69.9% recall, 77.5% mAP_{50} , and 36.1% mAP_{95} —surpassing the YOLOv8 baseline by 4.9% , 7.5% , 7.5% , and 3.6% , respectively. Notably, these improvements are attained using only 86% of the parameters of YOLOv8, underscoring the model's computational efficiency. Comparative evaluations with other state-of-the-art detectors confirm that YOLO-DTS offers significant advantages in PCB defect detection under complex imaging conditions.

Despite its strong performance, this study has limitations in terms of defect sample diversity and annotation cost, recognition under extreme clutter, and real-time inference speed. Future work will focus on integrating multi-scale attention and feature reconstruction mechanisms to improve discrimination of subtle and irregular defects; developing semi- and self-supervised learning protocols to leverage the abundant unlabeled data from production lines, thereby significantly reducing annotation dependence and addressing the inherent challenge of defect scarcity; and applying pruning and quantization techniques to further streamline the model for real-time industrial deployment without compromising accuracy.

CRedit authorship contribution statement

Yufeng Ou: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Chia-Hung Wang:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

The PKU-Market-PCB dataset used in this study is available at “<https://robotics.pkusz.edu.cn>”.

The DeepPCB dataset used in this study is available at “<https://github.com/Charmve/Surface-Defect-Detection/tree/master/DeepPCB>”.

The APSPC dataset used in this study is available at “<https://tianchi.aliyun.com/dataset/dataDetail?dataId=140666>”.

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